# 🚨 Kubernetes Error: Pod Cannot Access PersistentVolume

## 📌 What is a PersistentVolume (PV) and PersistentVolumeClaim (PVC)?

- PersistentVolume (PV) is a **storage resource** in the Kubernetes cluster.
- PersistentVolumeClaim (PVC) is a ***request*** for storage made by a pod.
- A pod can only use persistent storage when the PVC is successfully bound to a PV.
- If the PVC is not bound, the pod may remain in the **Pending** state.
- Example:
  - STATUS: **Pending** (or)
  - Warning **FailedScheduling** / (pod has unbound immediate PersistentVolumeClaims)..

## ✅ Check the PVC Status

```
kubectl get pvc
---
NAME        STATUS    VOLUME   CAPACITY   STORAGECLASS
mysql-pvc   Pending                                     #If the PVC status is Pending
```

## 🚀 PersistentVolume Troubleshooting Flow

```
Pod Cannot Access Storage
        ↓
kubectl get pvc
        ↓
PVC Status = Pending?
        ↓
kubectl describe pvc
        ↓
StorageClass Matches?
        ↓
PV Available?
        ↓
Access Modes Match?
        ↓
CSI Driver Running?
        ↓
Verify Pod Uses Correct PVC
```

PVC Bound Successfully

↓

PodStartsSuccessfully ✅

## ☸️ Kubernetes PV/PVC Issues — Common Reasons & Solutions

| 🚨 Cause | 📖 Description | 🛠️ Solution | 💻 Useful Commands |
|---|---|---|---|
| 📦 **PVC in Pending State** | No matching PersistentVolume (PV) is available | Create a matching PV or ensure dynamic provisioning is working | `kubectl get pvc` `kubectl describe pvc <pvc-name>` |
| 💾 **StorageClass Mismatch** | PVC requests a different `StorageClass` than the PV provides | Update the PVC or PV to use the same `StorageClass` | `kubectl get storageclass` |
| 📏 **Size Mismatch** | PVC requests more storage than the PV capacity | Increase the PV size or reduce the PVC request | `kubectl describe pv <pv-name>` |
| 🔄 **Access Mode Mismatch** | Access modes ( `ReadWriteOnce` , `ReadOnlyMany` , `ReadWriteMany`) do not match | Configure compatible access modes on the PV and PVC | `kubectl describe pv <pv-name>` |
| ❌ **PV Not Created** | No PersistentVolume exists (static provisioning) | Create a PV or enable dynamic provisioning with a `StorageClass` | `kubectl get pv` |
| ☁️ **CSI Driver Not Installed** | Storage driver (AWS EBS, Azure Disk, GCE PD, etc.) is missing or unhealthy | Install or verify the CSI driver and its controller/node Pods | `kubectl get csidrivers` |
| 🔗 **Incorrect PVC Reference** | Pod references the wrong PVC name | Update the Pod or Deployment to reference the correct PVC | `kubectl describe pod <pod-name>` |
| 🚫 **Volume Attachment Failure** | Cloud volume cannot attach to the node | Verify node health, cloud provider, and CSI controller logs | `kubectl describe pvc <pvc-name>` |

| 🚨 Cause | 📖 Description | 🛠️ Solution | 💻 Useful Commands |
| --- | --- | --- | --- |
| 🗑️ **Released or Failed PV** | PV is in `Released` or `Failed` state | Reclaim, recreate, or clean up the PV according to the reclaim policy | `kubectl get pv` |

## 🔍 Useful Troubleshooting Commands

| 💻 Command | 🎯 Purpose |
| --- | --- |
| `kubectl get pvc` | Check PVC status |
| `kubectl describe pvc <pvc-name>` | View PVC events and binding issues |
| `kubectl get pv` | List PersistentVolumes |
| `kubectl describe pv <pv-name>` | Inspect PV details |
| `kubectl get storageclass` | Verify StorageClass configuration |
| `kubectl get csidrivers` | Check installed CSI drivers |
| `kubectl get pods -n kube-system` | Verify CSI controller and node Pods |
| `kubectl describe pod <pod-name>` | Check volume mount errors |
| `kubectl describe pod <pod-name>` | Check pod volume-related errors |
| `kubectl get deployment <deployment-name> -o yaml` | Verify PVC references |
| `kubectl get events --sort-by=.metadata.creationTimestamp` | View recent cluster events |

## 🎯 Interview Answer

### Q: How do you troubleshoot PV/PVC issues in Kubernetes?

- I first check the PVC status using ***kubectl get pvc***. If the PVC is in the Pending state, I run ***kubectl describe pvc*** to identify the reason.
- Verify that the PVC is bound to a PV.
- Ensure the StorageClass matches between the PV and PVC.

- Confirm the **requested storage size** and **access modes** are compatible.
- Verify that the CSI driver is installed and healthy for dynamic provisioning.
- Check that the Pod references the correct PVC.
- Review Pod events for ***volume attachment*** or ***mount failures***.

## 🎯 Interview One-Liner

- ☸️ PV/PVC issues are commonly caused by ***pendingPVCs*** , ***StorageClass mismatches***, ***insufficient PVcapacity*** , ***incompatibleaccessmodes*** , ***missingPVs*** , ***unhealthy CSI drivers***, or ***incorrect PVC references*** .
- 🚀 The first troubleshooting step is to inspect the PVC with **kubectl describe pvc**, then verify the PV, StorageClass, CSI driver, and Pod volume configuration.